

Phase 39: Strict Subset Operator

The strict subset operator (psubset , $\subset$) represents proper subset relationships, where A is a subset of B AND A is not equal to B.

Distinction from subset ($\subseteq$): - A subset B means: every element of A is $\in$ B (may be equal) - A psubset B means: every element of A is $\in$ B AND $A \neq B$ (strict inequality)

Example 1 : Basic Strict Subset

$$\begin{array}{|l} A : \mathbb{P} \mathbb{N} \\ B : \mathbb{P} \mathbb{N} \\ \hline A = \{1, 2, 3\} \wedge B = \{1, 2, 3, 4, 5\} \wedge A \subset B \end{array}$$

Here A psubset B holds because all elements of A are in B, and A contains fewer elements than B.

Example 2 : Comparing subset and psubset

$$\begin{array}{|l} X : \mathbb{P} \mathbb{N} \\ Y : \mathbb{P} \mathbb{N} \\ W : \mathbb{P} \mathbb{N} \\ \hline X = \{1, 2\} \wedge Y = \{1, 2, 3\} \wedge W = \{1, 2\} \wedge X \subset Y \wedge X \subseteq W \end{array}$$

X psubset Y is true (X is strictly contained in Y) X subset W is true (X is a subset of W, including the case where $X = W$) X psubset W would be FALSE (they are equal sets)

Example 3 : Empty Set

$$\begin{array}{|l} \text{emptySet} : \mathbb{P} \mathbb{N} \\ \text{anySet} : \mathbb{P} \mathbb{N} \\ \hline \text{emptySet} = \{\} \wedge \text{anySet} = \{1, 2, 3\} \wedge \text{emptySet} \subset \text{anySet} \end{array}$$

The empty set is a strict subset of any non-empty set.

Example 4 : Transitive Property

If A psubset B $\wedge$ B psubset C, then A psubset C.

$$\begin{array}{|l} A1 : \mathbb{P} \mathbb{N} \\ B1 : \mathbb{P} \mathbb{N} \\ C1 : \mathbb{P} \mathbb{N} \\ \hline A1 = \{1\} \wedge B1 = \{1, 2\} \wedge C1 = \{1, 2, 3\} \wedge A1 \subset B1 \wedge B1 \subset C1 \wedge A1 \subset C1 \end{array}$$

Example 5 : Set Hierarchy

[*PersonSS*]

$$\begin{array}{|l} \text{students} : \mathbb{P} \text{PersonSS} \\ \text{grads} : \mathbb{P} \text{PersonSS} \\ \text{phds} : \mathbb{P} \text{PersonSS} \\ \hline \text{phds} \subset \text{grads} \wedge \text{grads} \subset \text{students} \end{array}$$

PhD students are a strict subset of graduate students, which are a strict subset of all students.